

Alfido Tech Internship

Task 2: Motor Control with PWM & Obstacle Detection

Name:

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Domain:

Embedded Systems

Tools Used:

- Tinkercad Circuit Simulator
 - Arduino IDE
-

1. Objective

The objective of this project is to control the speed of a DC motor using PWM (Pulse Width Modulation) and automatically stop the motor when an obstacle is detected using an ultrasonic sensor. The system demonstrates motor control and real-time obstacle detection using Arduino.

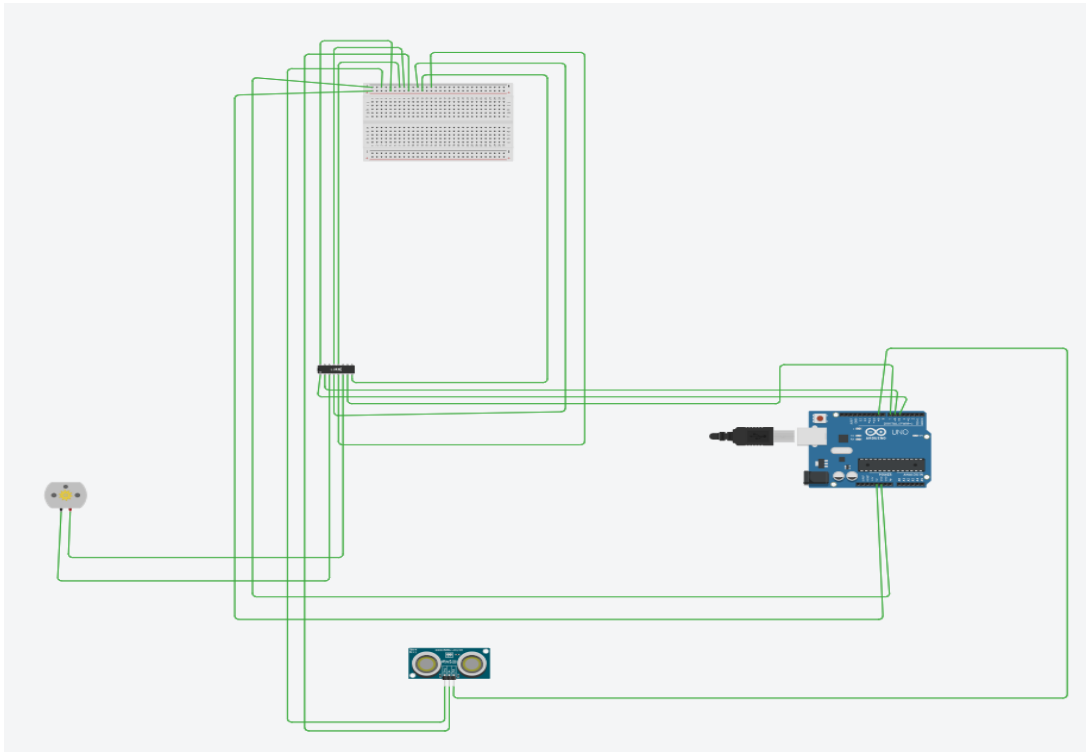
2. Components Used

- Arduino UNO
 - PING))) Ultrasonic Sensor
 - DC Motor
 - Motor Driver
 - Breadboard
 - Jumper Wires
-

3. Software Used

- Tinkercad Circuit Simulator
 - Arduino IDE
-

4. Circuit Diagram



5. Working Principle

The ultrasonic sensor continuously measures the distance of nearby objects and sends the distance data to the Arduino. The Arduino processes this information and controls the DC motor using PWM. When the detected distance is greater than 20 cm, the motor runs at the specified speed. If an obstacle is detected within 20 cm, the Arduino stops the motor automatically to prevent collision.

6. Arduino Code

```
const int pingPin = 9;

const int ENA = 5; // PWM Pin
const int IN1 = 6;
const int IN2 = 7;

long duration;
int distance;
```

```
void setup() {  
  Serial.begin(9600);  
  
  pinMode(ENA, OUTPUT);  
  pinMode(IN1, OUTPUT);  
  pinMode(IN2, OUTPUT);  
  
  // Motor Forward Direction  
  digitalWrite(IN1, HIGH);  
  digitalWrite(IN2, LOW);  
}  
  
void loop() {  
  
  // PING))) Sensor  
  pinMode(pingPin, OUTPUT);  
  digitalWrite(pingPin, LOW);  
  delayMicroseconds(2);  
  
  digitalWrite(pingPin, HIGH);  
  delayMicroseconds(5);  
  
  digitalWrite(pingPin, LOW);  
  
  pinMode(pingPin, INPUT);  
  duration = pulseIn(pingPin, HIGH);  
  
  distance = duration / 29 / 2;  
  
  Serial.print("Distance: ");  
  Serial.print(distance);  
  Serial.println(" cm");  
}
```

```

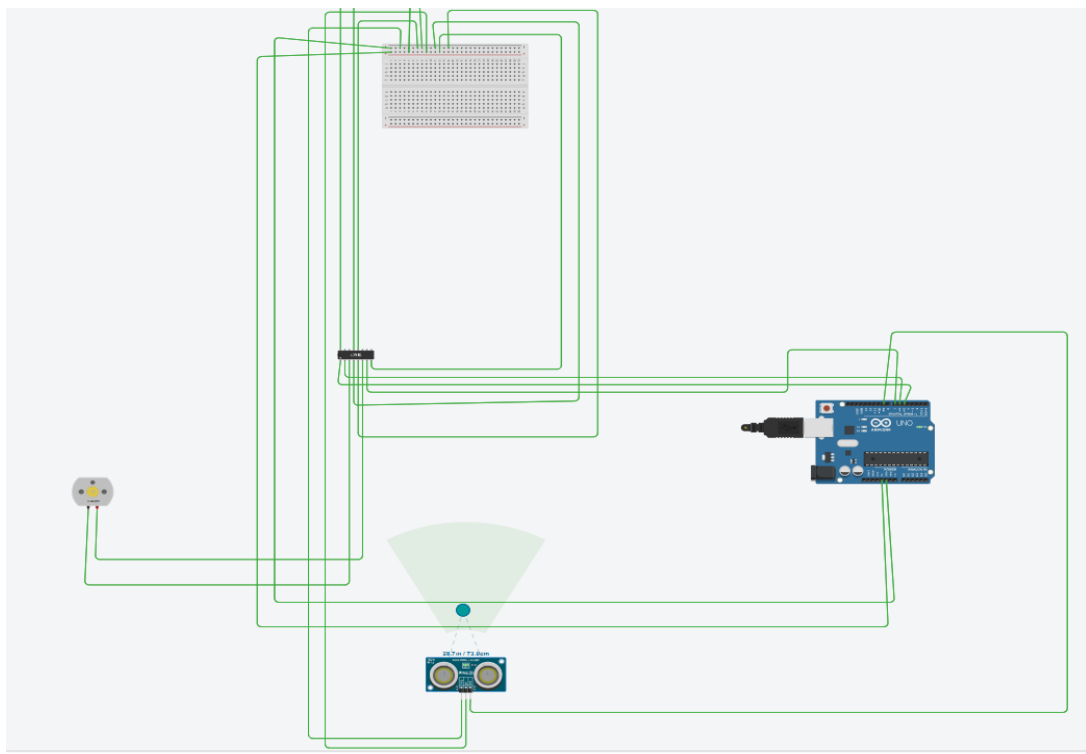
if (distance > 20) {
  analogWrite(ENA, 180);
  Serial.println("Motor Running");
}
else {
  analogWrite(ENA, 0);
  Serial.println("Motor Stopped");
}

delay(200);
}

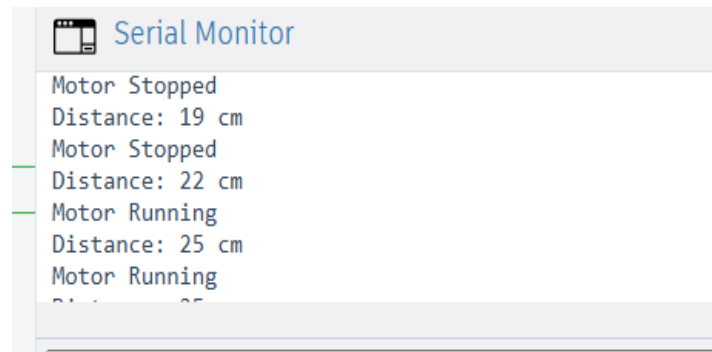
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7. Output Screenshots

Circuit Screenshot



Motor Running ,Stopped Output and serial monitor ouput



Code Screenshot

```
pwm_motor_control_amp_obstacle_detection1 | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Select Board

pwm_motor_control_amp_obstacle_detection1.ino

1  const int pingPin = 9;
2
3  const int ENA = 5;  // PWM Pin
4  const int IN1 = 6;
5  const int IN2 = 7;
6
7  long duration;
8  int distance;
9
10 void setup() {
11   Serial.begin(9600);
12
13   pinMode(ENA, OUTPUT);
14   pinMode(IN1, OUTPUT);
15   pinMode(IN2, OUTPUT);
16
17   // Motor Forward Direction
18   digitalWrite(IN1, HIGH);
19   digitalWrite(IN2, LOW);
20 }
21
22 void loop() {
23
24   // PING))) Sensor
25   pinMode(pingPin, OUTPUT);
26   digitalWrite(pingPin, LOW);
27   delayMicroseconds(2);
28
29   digitalWrite(pingPin, HIGH);
30   delayMicroseconds(5);
31
32   digitalWrite(pingPin, LOW);
33
34   pinMode(pingPin, INPUT);
35   duration = pulseIn(pingPin, HIGH);
36
37   distance = duration / 29 / 2;
38
39   Serial.print("Distance: ");
40   Serial.print(distance);
41   Serial.println(" cm");
42
43   if (distance > 20) {
44     analogWrite(ENA, 180);
45     Serial.println("Motor Running");
46   }
47   else {
48     analogWrite(ENA, 0);
49     Serial.println("Motor Stopped");
50   }
51
52   delay(200);
53 }
```

8. Results

The motor control system successfully operated using PWM speed control. The ultrasonic sensor accurately detected obstacles and automatically stopped the motor when the distance was less than or equal to 20 cm. The project worked successfully in the Tinkercad simulation environment.

9. Conclusion

This project demonstrates the fundamental concepts of embedded systems including PWM motor control, ultrasonic sensor interfacing, real-time distance measurement, and obstacle detection using Arduino. The system successfully controlled the motor and responded correctly to obstacle detection conditions.

10. GitHub Repository Link

<https://github.com/naveenapatel08-a11y/Task-2-Motor-Control-PWM-Obstacle-Detection>

11. Demo Video Link

<https://drive.google.com/file/d/1cpCSayH5pAbGUbkDLubsENofEc601znJ/view?usp=sharing>